Differential Equations for Master Integrals

A concise overview

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These notes summarize a longer treatment. The setting is the double-real channel of $pp \to h + X$ at NNLO: reverse unitarity turns its phase-space integrals into cut two-loop integrals, and integration-by-parts reduction leaves 347 master integrals depending on the Mandelstam ratios $v = -t/s$ and $w = -u/s$, whose physical region is the triangle $0 < v$, $0 < w$, $v + w < 1$, in $d = 4 - 2\varepsilon$ dimensions. The Born configuration sits on $v + w = 1$, so $1 - v - w \to 0$ is the soft limit and $v \to 0$, $w \to 0$ the collinear ones. The problem is the analytic evaluation of the masters: each as a Laurent series in ε whose coefficients are explicit transcendental functions with exact constants.

1 Differential equations

A derivative of a master with respect to v or w is again an integral of the same family, and integration by parts brings it back to the masters of the family and its subsectors. Each family therefore satisfies a closed first-order system

$$\mathrm{d}\vec{I} = A(v, w; \varepsilon)\, \vec{I}, \qquad A = A_v\, \mathrm{d}v + A_w\, \mathrm{d}w, \tag{1}$$

with rational matrices, subject to the integrability (flatness) condition $\mathrm{d}A = A \wedge A$, verified exactly. The families themselves are already fixed by the reduction: propagator sets are identified modulo redefinitions of the loop momenta that map cut lines to cut lines, the identification being found through a normal ordering of the Symanzik polynomials and then proved by the explicit momentum shift. Because differentiation never leaves a family, the 347 masters split into 91 independent systems, of dimensions between 2 and 47 (the dimensions sum to more than 347 because each system carries the subsector masters it couples to, counted once in the family that owns them). Ordered by sectors — a sector is the set of propagators actually present — each connection is block lower triangular: diagonal blocks couple the masters of one sector group, and the entries below the diagonal feed subsector solutions upward.

2 Equivalence of blocks

A *block* is a strongly connected group of masters, a diagonal block of the triangular structure; the 91 systems contain 1119 of them. Two blocks are *equivalent* when their connections coincide after a basis permutation, optionally composed with the crossing involution $v \leftrightarrow w$ — an exact matrix identity, never an analogy. Only 173 equivalence classes occur, because the same subtopologies recur inside many parent families. The analysis is therefore done once for each class; a family is recovered afterwards from the classes it contains, and the identification is checked again at that point.

3 The canonical form

A system is in *canonical form* when

$$\mathrm{d}\vec{F} = \varepsilon \sum_a R_a \,\mathrm{d}\log\varphi_a(v,w)\; \vec{F}, \tag{2}$$

with constant residue matrices R_a and ε-free *letters* φ_a (the *alphabet*). The Laurent expansion then decouples order by order, $\mathrm{d}\vec{F}^{(k)} = \big(\sum_a R_a \,\mathrm{d}\log\varphi_a\big)\vec{F}^{(k-1)}$: each order is a quadrature of the previous one, and the solution is a path-ordered exponential whose order-ε^k term is a sum of k-fold iterated integrals over the alphabet — generalized polylogarithms of uniform weight k. Take the alphabet $\{w, 1-w\}$ with a nilpotent coupling between the two components: the first order in ε produces $\log(1-w)$ and the second $\mathrm{Li}_2(w)$ — the dilogarithm appears as the twofold integral of the word $(w,\,1{-}w)$, and nothing more elaborate is involved. All singular behaviour is manifest: the connection has simple poles exactly at the letters' zeros, and the residue matrix there prescribes the local solutions $w^{\,n+a\varepsilon}(\log w)^j$ — exponents from ε times its eigenvalues, logarithms from its Jordan structure — known before any integration.

4 Construction of the transformation

Under a change of basis $\vec{I} = T\vec{F}$ the connection transforms as $A \mapsto T^{-1}AT - T^{-1}\mathrm{d}T$, and the triangular structure decides the order of the work. A diagonal block never feels what lies below it, so each block is brought to canonical form on its own — once per class, by rational-ansatz methods or by Fuchsian reduction with balance transformations, a candidate counting as the canonical form only when the transformed block equals $\varepsilon\sum_a R_a \,\mathrm{d}\log\varphi_a$ identically. With the diagonal canonical, the entries below it yield to unipotent transformations $T = 1 + D$ (so $D^2 = 0$): writing εe_k and εc_j for the canonical diagonal blocks of row k and column j, and $\bar{b}_{kj}$ for the coupling to be removed (including the feed-through of previously completed entries of the same row), the unknown block D_{kj} and the constant residues K_a of the completed entry satisfy the *linear* equation

$$\mathrm{d}D_{kj} = \varepsilon(e_k D_{kj} - D_{kj}c_j) + \bar{b}_{kj} - \varepsilon\textstyle\sum_a K_a \,\mathrm{d}\log\varphi_a. \tag{3}$$

Being linear, it is solved by an ansatz over the letters with undetermined polynomial numerators, evaluated at many rational kinematic and regulator values modulo large primes; the exact rational solution — a rational function of ε as well — is reassembled by the Chinese remainder theorem and rational reconstruction, and proved by exact substitution. Where the solution is not unique, the representative with vanishing top-degree numerator coefficients is chosen: all representatives are correct, but a low-degree one keeps the degrees of the later rows from compounding. What survives is a dependence on ε that is constant in the kinematics — a residue of the ε-dependent normalization of the basis — and it is removed last, by a conjugation $T(\varepsilon)$ found from the connection at a few rational points and proved exactly.

5 Square roots and their rationalization

Degenerate configurations of the three-particle final state lie on the vanishing of the Källén polynomial $\lambda_1 = (1-v-w)^2 - 4vw$, whose zero locus $\sqrt{v}+\sqrt{w} = 1$ is a pseudo-threshold *inside* the physical triangle; alphabets of many blocks contain $\sqrt{\lambda_1}$-type letters. Six radicands occur in all; the involution $v \leftrightarrow w$ exchanges two pairs of them and fixes the other two. The *root content* of a family is the number of independent square roots in its alphabet, counted so that $\sqrt{q_1}$, $\sqrt{q_2}$, $\sqrt{q_1 q_2}$ are two roots, not three: most families have none, a third have one, a dozen have two, and three families have three. A single root is removed by a *rationalizing substitution* — for example $v = xy$, $w = (1-x)(1-y)$ gives $\lambda_1 = (x-y)^2$, so $\sqrt{\lambda_1} = x - y$ and the alphabet becomes polynomial in (x, y); such substitutions come from the classical parametrization of a conic by lines through a rational point. Two roots are removed by iterating the construction inside the first substitution. For three roots no rationalizing substitution exists — the joint covering surface resolves to a K3 surface, which is not unirational — and one computes instead in the extension $\mathbb{Q}(v, w, \varepsilon)[r_1, r_2, r_3]/(r_i^2 - q_i)$, whose Galois group $(\mathbb{Z}/2)^3$ acts by sign changes: linear problems are posed on the eight graded components (ordinary rational functions), and the eight sign embeddings must and do recombine consistently.

6 Paths and polylogarithms

With the canonical form established, evaluation means choosing a base point in the physical triangle and a path composed of segments on which every letter is linear in the path parameter; along such segments the iterated integrals are the generalized polylogarithms $G(a_1, \ldots, a_k; t)$. A letter quadratic on a segment stays quadratic on every sub-segment, so one either chooses the segments differently — in practice parallel to the axes, where the bilinear letters of the rationalizing substitutions become linear again — or admits the quadratic's linear factors as letters with algebraic constants. Flatness makes the result depend only on the endpoint and the homotopy class of the path; letters' zeros are crossed only as deliberate analytic continuation. The order in ε carried for each master is fixed in advance by where it enters the cross section: most masters are needed only to their finite order, a large minority one order deeper, and a handful deeper still.

7 Boundary conditions

The general solution leaves one constant vector per family per order, and almost all of it is fixed without evaluating any new integral. The most productive condition is the exclusion of local behaviour: at $v \to 0$, $w \to 0$ and $1 - v - w \to 0$ the integral representation admits only exponents $n + a\varepsilon$ with a from a short list fixed by the analysis of momentum regions (hard, soft, collinear), while the equation offers exponents from the residue eigenvalues — and every offered-but-forbidden mode annihilates one constant. The rest follows from elementary normalizations (the phase-space volume is a closed ratio of Γ-functions and fixes the lowest sector wherever it appears), from inheritance (families are treated in subsector order, and every already-known subsector master imposes its expansion wherever it recurs), and from finiteness at letters' zeros interior to the region of convergence. The number of genuinely undetermined constants of a block is the kernel dimension of its constraint system, known before any evaluation; after identification across blocks, the few surviving *boundary periods* are values of cut phase-space integrals at ordered limits — ζ-values and kin — established by exact derivations, with numerics as corroboration only.

8 Edges of phase space

Near a letter's zero the local solutions are $u^{n+a\varepsilon}(\log u)^j$, with all data read from the residue matrix. At the endpoints the cross section requires the distributional identity

$$(1-w)^{-1+a\varepsilon} = \frac{1}{a\varepsilon}\,\delta(1-w) + \sum_{k\geq 0} \frac{(a\varepsilon)^k}{k!}\left[\frac{\log^k(1-w)}{1-w}\right]_+, \tag{4}$$

where $[\,\cdot\,]_+$ subtracts the value of the test function at the endpoint; the identity exists only for the unexpanded power, since expanding the master in ε first produces non-integrable logarithms and destroys the δ-content. The masters therefore carry their exact local mode data alongside the polylogarithmic expansion, and the identity is applied before the final expansion in ε.

9 Search and proof

Nothing depends on how a transformation was found: an ansatz, a fit, arithmetic modulo a prime produce candidates, and what makes a candidate a theorem is an exact identity — flatness for a connection, reconstruction from constant residues for a canonical form, the transformation identity for an assembled family, a numerics-free derivation for a boundary period. At the time of writing all but the three three-root families carry canonical forms established in this way; the three-root families await the Galois-algebraic treatment, and the higher boundary periods and the endpoint analysis of the full inventory are open campaigns. The method is complete; the calculation, like every large calculation, is a living object.